The leading constant in polynomial evaluation

over algebraically closed fields

Fix a polynomial $p \in K[x]$. Starting with 1 and x, we may take linear combinations of anything we have already computed, at no cost. We pay one for each multiplication of two nonconstant polynomials. The coefficients of p are known when we choose the computation. How many multiplications suffice for the hardest polynomial of degree at most n?

Write $M_K(n)$ for this number. Paterson and Stockmeyer's algorithm gives $M_K(n) \le \sqrt{2n} + O(\log n)$ [3]. Over an infinite field, some polynomials require at least $\sqrt{n} - O(1)$ multiplications [1, Theorem 2.2].

Theorem. *Over every algebraically closed field K,*

$$M_K(n) \;\le\; \sqrt{n} + O\left(\frac{\sqrt{n}}{\log n}\right).$$

In particular, $M_K(n)/\sqrt{n} \to 1$.

What one multiplication can do

Suppose we have made several products and are about to make another. We get to choose a linear combination of the available polynomials for *each* of its two inputs. As the computation grows, these choices accumulate. With s products there are on the order of s^2 scalars to choose. Could we use them to prescribe all the coefficients of a polynomial of degree $s^2 - 1$?[1]

Here is a small example. Suppose that x^4, x^{10}, x^{14} are already available. With one more multiplication we can form

$$H = (x^{10} + a + bx)(x^4 + c + dx) - x^{14}.$$

The subtraction removes the product at $a = b = c = d = 0$. Turning on a alone gives ax^4. Turning on b, c, or d alone gives bx^5, cx^{10}, or dx^{11}. With all four present we get

$$H = \underbrace{ax^4 + bx^5 + cx^{10} + dx^{11}}_{\text{linear in the four scalars}} + (a + bx)(c + dx).$$

The four independent changes we can see here occur at rather inconvenient degrees. But suppose that what we want from H is its remainder modulo $x^4 - 1$. In that remainder, x^4, x^5, x^{10}, x^{11} become $1, x, x^2, x^3$, in order. To first order in the four scalars, we can therefore prescribe every coefficient of a cubic.

To use many copies of this idea, we ask each output for a different remainder of p. Remainders modulo pairwise coprime polynomials determine p as soon as their degrees add up to more than $\deg p$. We will make the inputs adjustable enough to prescribe each remainder to first order. Algebraic closedness will let us account for the mixed terms and attain all the desired remainders up to one common scalar. What remains is to recover p cheaply enough to preserve the gain.

[1] Over $\mathbb{C}$, Jarlebring and Lorentzon prove that the family computed by $s \ge 3$ products has dimension s^2 [2]. Its polynomials can have degree as large as 2^s.

1 How to keep the changes distinct

In the example, the term cx^{10} falls four degrees short of the original product x^{14}. Multiply both by x^m and the new term cx^{10+m} still falls four degrees short of x^{14+m}. We will use this *loss in degree* to recognize a change many multiplications after it was made.

For the moment, write the variable as z. Consider the sequence of reference products

$$z \cdot z^2 = z^3, \qquad z^3 \cdot z^6 = z^9, \qquad z^9 \cdot z^{18} = z^{27}, \quad \ldots$$

The first input at each step is the previous result. The second is a separately prepared monomial. At a step with inputs z^e and z^{2e}, add a scalar multiple of some earlier power z^f to one input. If we add it to the first input, the new term has exponent $f + 2e$ and hence falls $e - f$ short of $3e$. If we add it to the second, the loss is $2e - f$. Later steps preserve the loss. We use powers of three because these differences remember both *when* the change was made and *which* power was used.

Why stop this sequence after a few steps? After L steps its reference exponent is 3^L. That growth matters when we turn a remainder back into a polynomial. If a computed expression H agrees with p modulo Q, we have $H = p + QT$. We can afford to evaluate the quotient T separately if its degree is small. A sequence of length $\sqrt{n}$ would give us no such control.

Instead we run many short sequences side by side. At each step we can still use earlier results from *all* the sequences as inputs. This keeps the number of scalar choices large even though the individual sequences are short.

Here is the circuit. Take b sequences, each of length L, with $3^L \le b$. Number them $0, \ldots, b-1$. Distinguish their starting values by a small power of x, setting

$$z = x^{2b}, \qquad G_{i,0} = x^i z \quad (0 \le i < b).$$

The spacing $2b$ leaves room for two of the small exponents to add without changing the exponent of z. At step $j = 1, \ldots, L$ of sequence i, make the single product

$$G_{i,j} = (G_{i,j-1} + A_{i,j})(z^{2 \cdot 3^{j-1}} + B_{i,j}). \tag{1}$$

Both $A_{i,j}$ and $B_{i,j}$ are free linear combinations of earlier results $G_{a,t}$, with a separate scalar for each summand. In $A_{i,j}$ we allow $0 \le t < j - 1$ and in $B_{i,j}$ we allow $0 \le t < j$. In both cases a runs through all b sequences. Thus at the first step $A_{i,1} = 0$, while $B_{i,1}$ can use all b starting values. The strict inequality in the first input ensures that every substitution incurs a positive loss.

When all these scalars are zero, each sequence simply multiplies by the indicated powers of z, giving $G_{i,j} = x^i z^{3^j}$. Subtract the known reference value at the end, and call the result $Y_i = G_{i,L} - x^i z^{3^L}$. All b outputs now start at zero.

2 Reading a change from its exponent

Choose one of the scalars in (1), say the coefficient of $G_{a,t}$ at step j of sequence i. Keep all other scalars zero. What reaches the outputs when we turn this one on? At the chosen product the new term is $x^a z^{3^t}$ times the unchanged input. From there it travels to the end of sequence i, multiplied at every later step by the same monomial as the reference term. It reaches no other output, since every coefficient that could carry it to another sequence is zero.

More explicitly, the coefficient of our chosen scalar in Y_i is a monomial $x^u z^{3^L-d}$. Its small exponent u and its loss d are

input changed	allowed t	u	d
first	$0 \le t < j-1$	a	$3^{j-1} - 3^t$
second	$0 \le t < j$	$i+a$	$2 \cdot 3^{j-1} - 3^t$.

For example, at step $j = 3$ the reference inputs have z-exponents 9 and 18. Using a term of exponent 3 in their place gives losses 6 and 15. From either loss we can recover the source exponent 3 as the largest power of three that divides it. Adding this 3 back tells us which input was changed.

The same decoding works in general. Since $u < 2b$, the ordinary degree of the monomial determines u and d. Write $d = c3^{j-1} - 3^t$ where $c = 1$ or 2. The largest power of three dividing d is 3^t. For $t < j-1$ this follows by factoring out 3^t. For $t = j-1$ only the second input is allowed and $d = 3^t$. Adding back 3^t recovers $c3^{j-1}$, hence both j and the input. Finally u gives a, since we know the output i. This arithmetic takes place in the integer exponents and works in every characteristic.

At step j of a given sequence there are $b(j-1)$ scalar choices in the first input and bj in the second. We have therefore produced

$$k = b \sum_{j=1}^{L} (2j-1) = bL^2$$

distinct monomials in the linear part of each output, one for each scalar used in that sequence. Across all b outputs there are $bk = (bL)^2$ such choices for our bL products.

We will also need a bound on the degrees when *all* scalars are in use. It is

$$\deg G_{i,j} \le 2b\,3^j + b - 1, \qquad \deg Y_i \le D := 2b\,3^L + b - 1. \tag{2}$$

Indeed, the first bound holds at $j = 0$. If it holds through step $j-1$, the first input has degree at most $2b\,3^{j-1} + b - 1$. In the second input every earlier result has degree at most this much, and the fixed monomial has the larger degree $4b\,3^{j-1}$. Adding the input bounds gives (2).

3 Prescribing the remainders

We now have k distinct monomials at each output. In the opening example reduction modulo $x^4 - 1$ turned four particular monomials into a basis. For an arbitrary list of k monomials there is a convenient way to find a modulus that does the same job. A remainder of degree less than k is determined by its values at k distinct points. Choose these points in a geometric progression. Then *each monomial*, read along the progression, is itself a geometric progression, with a different ratio for each exponent.

In detail, choose $\omega \in K^\times$ such that $1, \omega, \ldots, \omega^D$ are distinct. This excludes only the roots of $X^d - 1$ for $1 \le d \le D$, a finite set in the infinite field K. Set

$$R(x) = \prod_{h=0}^{k-1} (x - \omega^h).$$

The values of $x^{e_1}, \ldots, x^{e_k}$ at these points form the Vandermonde matrix $\left((\omega^{e_a})^h\right)_{0 \le h < k,\, 1 \le a \le k}$. It is invertible for any distinct e_a between 0 and D. Thus at each output the k monomials form a basis modulo R.

If we took every remainder modulo this same R, the outputs would all be specifying the same k coefficients of p. To see how to give them separate jobs, consider two moduli R and $R - 1$. The expression $(1 - R)U + RV$ gives U modulo the first and V modulo the second. With moduli of the form $R - \xi$, each new value of ξ gives us another k coefficients, and we can join all the outputs by interpolation in R.

We must check that the monomials still form a basis after replacing R by $R - \xi$. For a given output, the coefficients of its k monomials reduced modulo $R - \xi$ are polynomials in ξ, by division by a monic polynomial. Their determinant is nonzero at $\xi = 0$, so only finitely many ξ spoil the basis. Choose distinct $\xi_0, \ldots, \xi_{b-1}$ so that the monomials at output i form a basis modulo $Q_i = R - \xi_i$.

Let θ denote the list of all bk adjustable scalars. In the map

$$\Phi : \theta \longmapsto \left(Y_0(\theta) \bmod Q_0, \ldots, Y_{b-1}(\theta) \bmod Q_{b-1}\right)$$

both source and target have bk scalar coordinates. The map is polynomial in those coordinates and $\Phi(0) = 0$. Its linear part is invertible because the scalars from each sequence give a basis in its own component and zero in every other component. The following fact converts that calculation at zero into a statement about every target.

Lemma. *Let $\Phi : K^r \to K^r$ be a polynomial map over an algebraically closed field. If $\Phi(0) = 0$ and its linear part J at zero is invertible, then for every $v \in K^r$ the image of Φ contains λv for some $\lambda \in K^\times$.*

Proof. First solve $\Phi(w(t)) = tv$ in formal power series, starting with $w(0) = 0$. If we have solved through degree $h - 1$, adding $t^h c$ to $w(t)$ changes the coefficient of t^h by Jc and preserves all lower coefficients. Invertibility of J lets us solve successively to every order.

The equations $\Phi(w) = \lambda v$, $\lambda y = 1$ therefore have a solution over $K((t))$, with $\lambda = t$ and $y = t^{-1}$. Their ideal over K is proper. By the weak Nullstellensatz they have a solution over K. $\square$

4 From remainders to the polynomial

Let $\deg p < bk$. Apply the lemma to the list of remainders of p modulo the Q_i. Some choice of the scalars gives, simultaneously,

$$Y_i \equiv \lambda p \pmod{R - \xi_i} \qquad (0 \le i < b)$$

for a single $\lambda \neq 0$. How do we turn these congruences into a circuit for p?

Modulo $R - \xi_j$ the polynomial $R(x)$ acts as the scalar ξ_j. Thus we can select any one of the remainders by ordinary interpolation. Let ℓ_i be the polynomial of degree less than b that is 1 at ξ_i and 0 at all the other ξ_j. Then the sum

$$F = \sum_{i=0}^{b-1} \ell_i(R)\, Y_i$$

agrees with Y_j modulo $R - \xi_j$ for every j. Since the moduli are pairwise coprime, $F - \lambda p$ is divisible by their product Q. Write $F - \lambda p = QT$. Now $\deg Q = bk$ and $\deg F \le (b-1)k + D$ by (2). Consequently $\deg T \le D$ if $T \neq 0$. Its coefficients, like those of p, are known when the circuit is chosen. Our answer is the polynomial identity

$$p = \lambda^{-1}(F - QT).$$

It remains to count the products outside the bL steps of (1). The shared monomials are inexpensive. Prepare $1, x, \dots, x^{2b}$ and then repeatedly square and multiply to pass from z^{3^j} to $z^{3^{j+1}}$. This also supplies the fixed second inputs in (1). Forming the b starting values and the b reference terms to be subtracted costs at most $2b$ more products. This preparation costs $O(b + L)$ products altogether.

For the interpolation and correction, recall that a known polynomial of degree $d \ge 1$ can be evaluated with $O(\sqrt{d})$ products. Prepare the powers through x^h, with $h = \lceil \sqrt{d} \rceil$, and use Horner's rule in x^h on blocks of h coefficients. Here $\deg R = k = bL^2$ and $\deg T \le D = 2b\,3^L + b - 1$. Both are $O(b^2)$ since $3^L \le b$, so R and T cost only $O(b)$ products. Prepare $1, R, \dots, R^b$ with another $O(b)$ products. These give every $\ell_i(R)$ and $Q = \prod_i (R - \xi_i)$ by free linear combinations. Forming F and QT takes at most $b + 1$ more products. Thus

$$M_K\big((bL)^2 - 1\big) \le bL + O(b) \qquad (3^L \le b).$$

For sufficiently large n, take $L = \lfloor \frac{1}{4} \log_3(n+1) \rfloor$ and $b = \lceil \sqrt{n+1}/L \rceil$. Then $3^L \le b$ and $\sqrt{n+1} \le bL < \sqrt{n+1} + L$. Since $b = O(\sqrt{n}/\log n)$, the theorem follows.

References

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